Information Visualization

Wrap up!

Lesson 10

Marilena Daquino
Assistant Professor

Department of
Classical Philology
and Italian Studies

marilena.daquino2@unibo.it

Table of contents

01

Publication

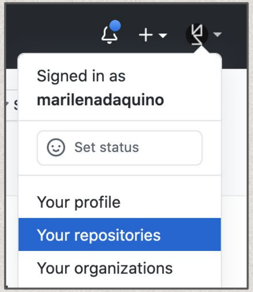
How to prepare your website for Github pages

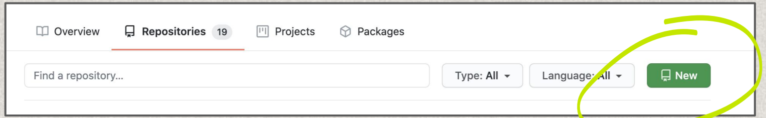
GitHub repo

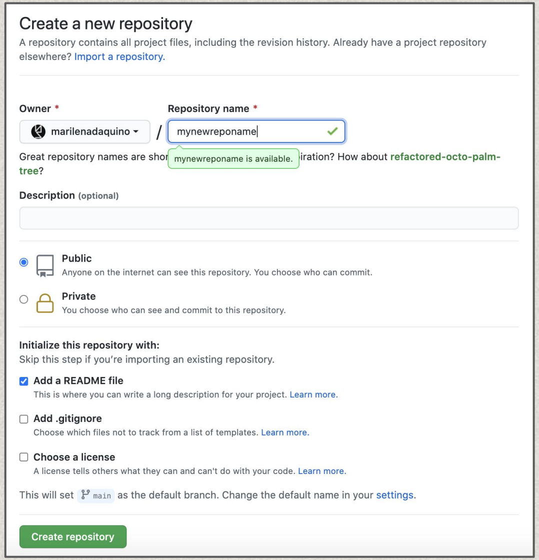
01

Publication

Create a repo







Must be public

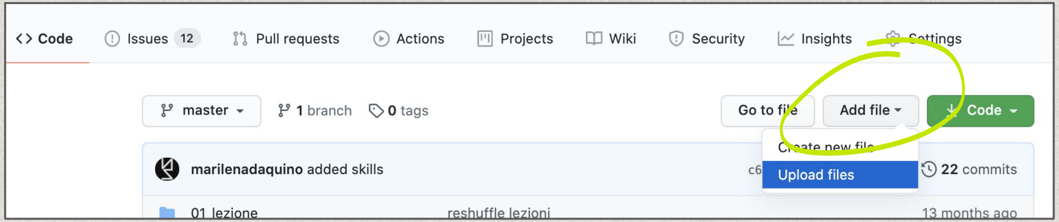
Must include a README file

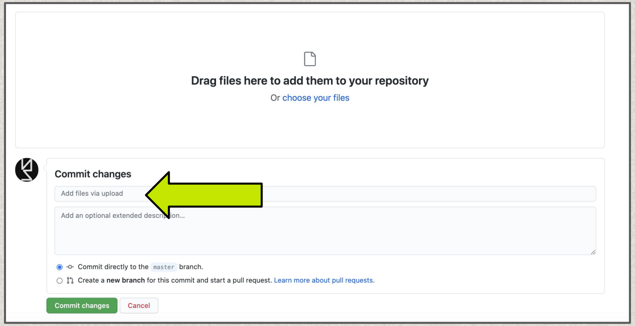
GitHub upload

01

Publication

Upload via GUI or client

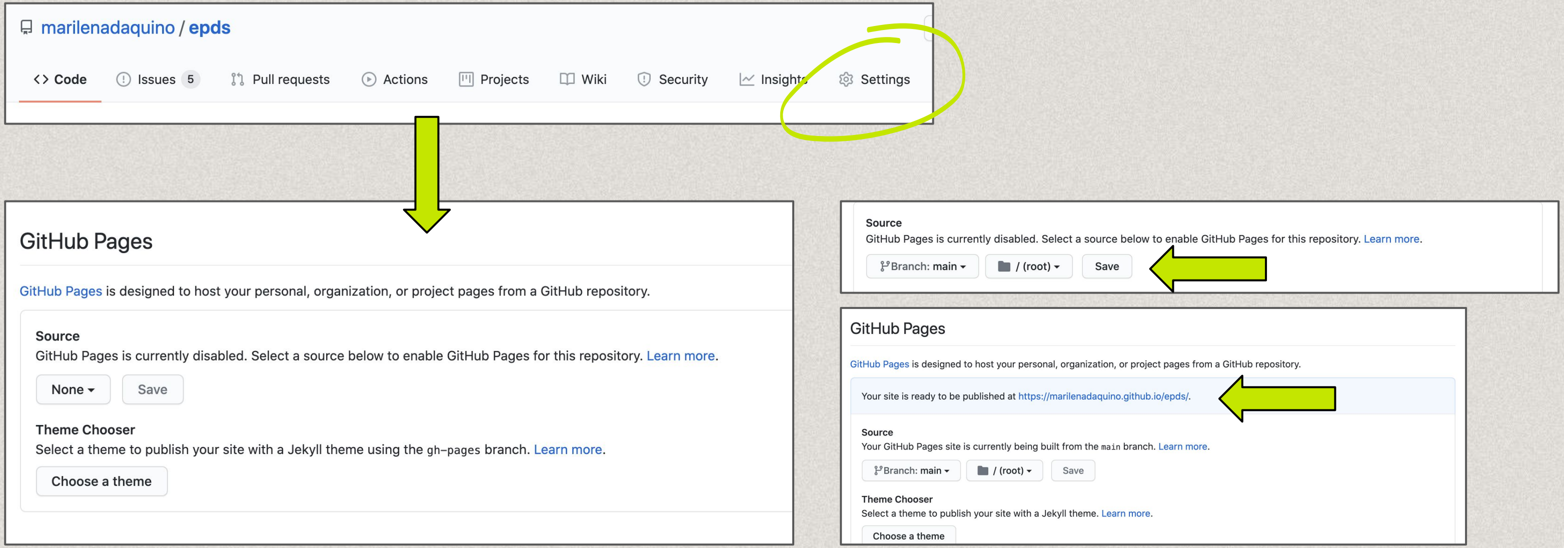




Short, meaningful commits

GitHub pages

Publish your website



What to upload

Data

OPTIONAL

Raw data used for the analysis if applicable - pay attention to licenses

MANDATORY

The data resulting from the analysis (you will use these for the website)

Notebook

One file .ipynb including: data integration, data cleaning, and data manipulation (profiling, exploratory analysis)

Website

HTML/CSS/JS files

The file index.html goes in the root folder.

README file

Title, links, roles (email?), licences

Edit file | Preview changes | Spaces | 2 | Soft wrap

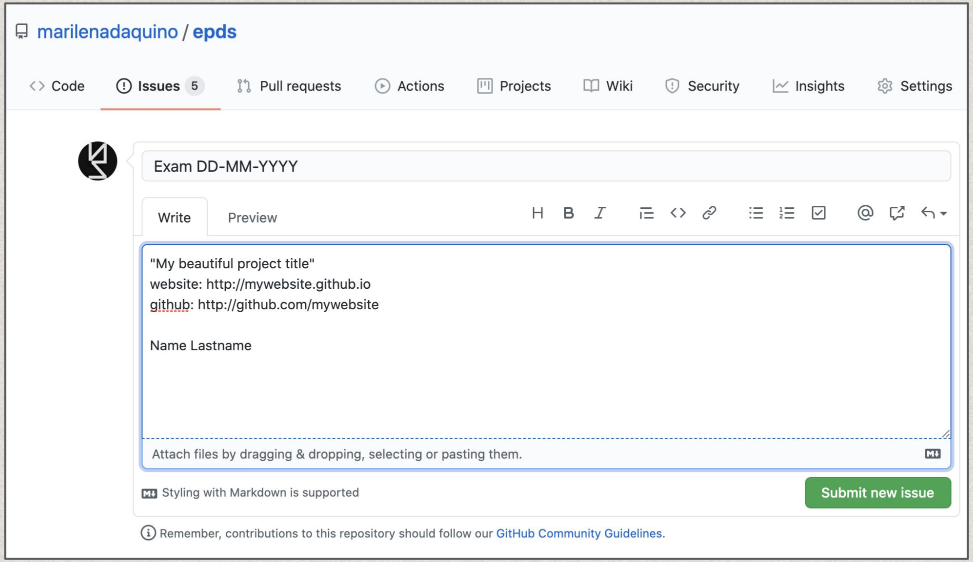
```
# My project title

[Project Website](http://website.github.io)

Jupyter notebook on [![Binder](https://mybinder.org/badge_logo.svg)](https://mybinder.org/v2/gh/your_account/your_repo/main)

Credits:

 * Name Lastname: role (e.g. data analysis, literature review)
 * Name1 Lastname1: role (e.g. web development, design)
 * Name2 Lastname2: role (e.g. data query, data integration)

License:

 * artchives data are released under a [CC0 license](https://creativecommons.org/share-your-work/public-domain/cc0/)
 * integrated data (if applicable) are reused and released under the original license (specify which one, e.g. Wikidata is CC0!)
 * code and final web application are released under [CC-BY license](https://creativecommons.org/licenses/by/4.0/)
```

optional

The day before the exam

At noon, not midnight…

Open an issue on the COURSE repo



1. Check if an issue with the exam date already exists
   a. If so, comment the issue
   b. If not, open a new issue with the date of the exam
2. Project title, URL of repo and website, names of group members

02

Assessment

How I grade exams

You remember this?

WHAT

Data

Correct and efficient SPARQL queries

Correct final CSV/JSON data

Data integration with multiple sources

Questions

Soundness and usefulness of research questions

Use of adequate visualizations

Graphic skills 30

Presentation

Clarity of the presentation during the exam 24

Ability to summarise complex issues 28

Surprise me! 30L

My table

Assessment

Date	Candidate	Group	Role	Online project URL	Repository
02/04/2021	[redacted]	☑	data visualization	https://hu	https://githu
02/04/2021	[redacted]	☑	data analysis and \	https://hu	https://githu
02/04/2021	[redacted]	☑	data analysis	https://digitalstory33.g	

Clear RQ	Clear goal	Automatic data collection	Data prep.	SPARQL queries	Data integration	Quality of EDA in Jupyter	Quality of Jupyter Source code	Quality of Web communication skills	Quality of graphic skills	Quality of results	Quality of Website Source code
☑	☑	☑	☑	☑	☐	☑	☑	☐	☐	☑	☐
☐	☐	☑	☐	☐	☑	☑	☑	☐	☑	☐	☐
☑	☑	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Usage of credits and licenses	Good presentation skills	Grade	Lode	Notes	Average
☐	☑	30	Lode		29.37777778
☐	☑	30	Lode		30/30L
☐	☐	30	Lode		28/30
☐	☐	28		wrong use of pie	24/30

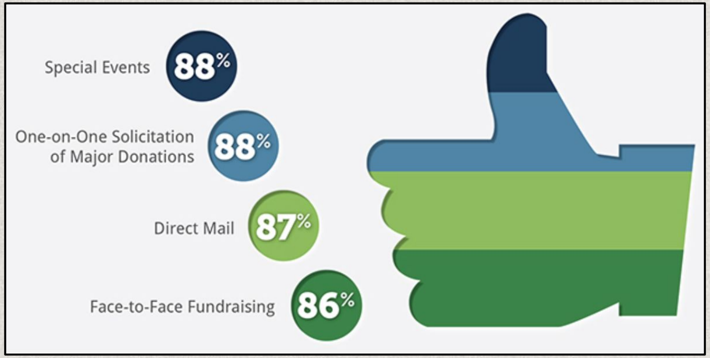
Your table

Self-assessment

Understanding of graphs/charts purpose	Usage of correct charts to represent data at hand
Selection of graphs/charts	Given the data at hand, usage of the most **effective** chart
Appropriateness of contents	Right amount of background/context/data/methodology (and in scope)
Strategic use of communication aspects	Titles, labels, explanations, visual clutter, color, white space
Overall design	No misspellings, no errors in data analysis. Correct order of contents. Clear message/action. Optimization of explanatory graphs / texts (summarisation)
Reliability of results*	*The conclusions make sense in the reality.*

** Make biases and limitations clear*

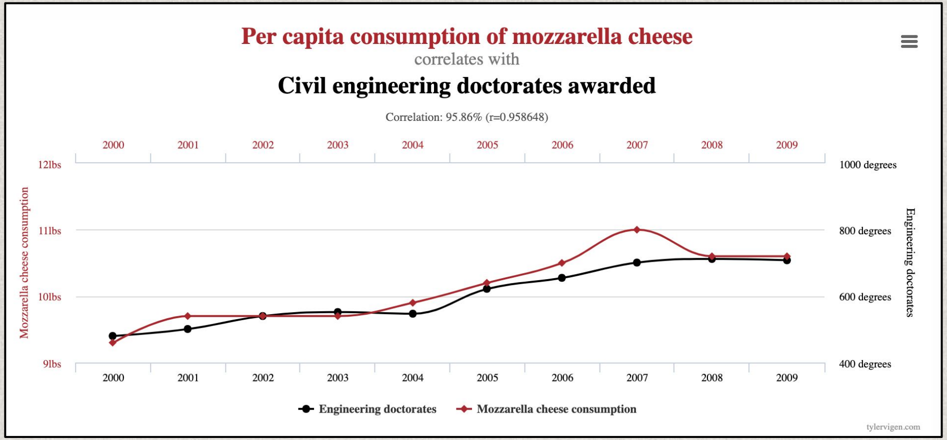
Some common mistake



ehm...

- The shape is not adequate to show proportions
- Percentages look weird right?

Some common mistake



ehm...

- Although funny, does not seem to be a credible result

** Make biases and limitations clear*

Some common mistake

ehm...

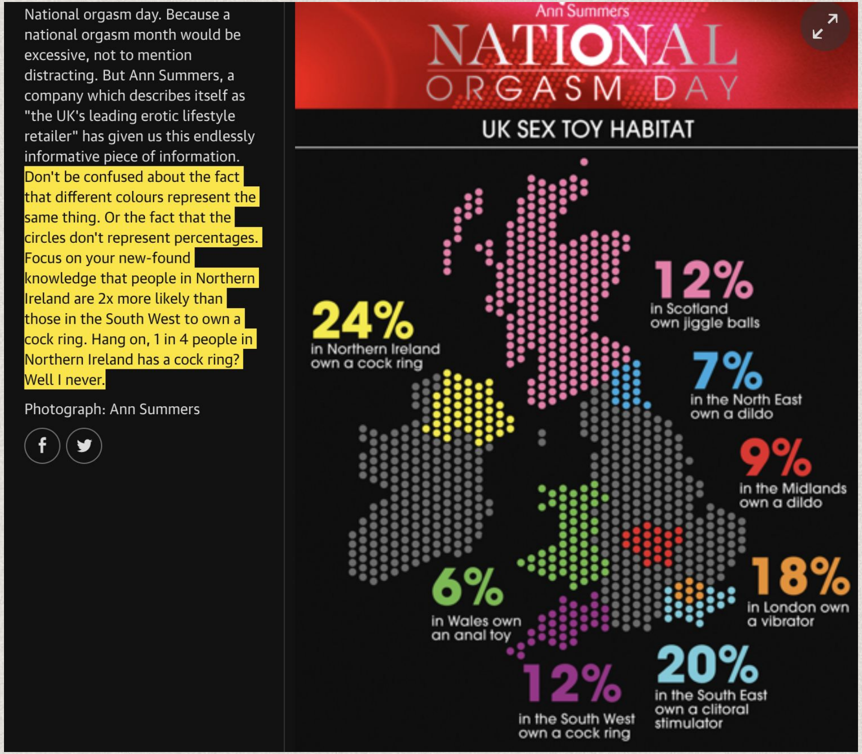
- We can confidently say that scaling is not this guy's thing

** Make biases and limitations clear*

Some common mistake

02

Assessment



ehm...

- Colors and shapes are used to differentiate areas, not data
- Percentages seem a little odd

** Make biases and limitations clear*

03

Assignments

Solutions and results of questionnaires

Q1

What question(s) are you trying to solve (or prove wrong)?

Some examples

Show **trends** in the study of artistic periods

Correlation between studied periods and **countries** (of historians)

Can we predict historians' interests based on their country/profession?

What kind of data do you have and how do you treat different types?

Categorical data (content, creators names, countries)

Numerical data (extent, number of items in the archive)

Ordinal data (dates)

What's missing from the data and how do you deal with it?

Example

Birth and death dates are missing -> can be integrated using external data sources (e.g. Wikidata)

Where are the outliers and why should you care?

Example

Scope and extent of collections (e.g. 290K photos in Zeri, broad timespan)

Outliers can affect results. We can apply some normalisation on data (e.g. using percentages rather than discrete numbers)

How can you add, change or remove features to get more out of your data?

Example

Extract knowledge from unstructured texts like bios or archive descriptions.

Topics of collections could be grouped in broader categories (e.g. artistic movements, periods)

Q2

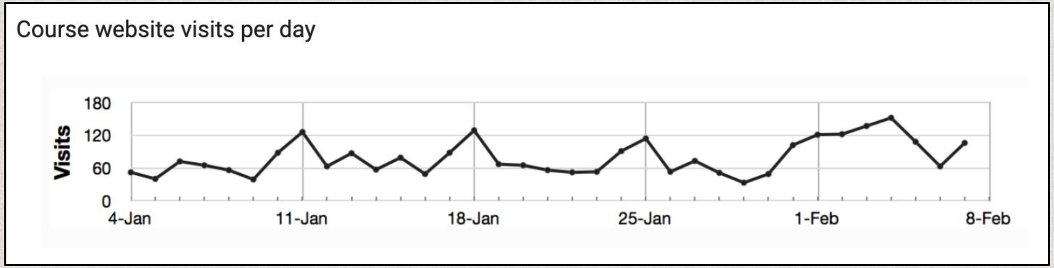
03

Assignments

Answers

It's a **graph**, specifically a **line graph** where there are **2** variables plotted, namely: **ordinal** values on the x axis, and **numeric** values on the y axis. Line graphs **order in 2D** data points.

The chart respects Tufte's principles: minimize data-ink ratio and chart junk, optimal scale and ratio.



Vertical lines represent **1-week** time spans.

Regularly, **every beginning (or end) of week there is a peak of visits**. The highest peak is in the middle of the first week of February though. Peaks may represent **weekly assignments** submitted by students, and the last peak may be the **final exam**.

Q3

Retrieve the list of (unique) historians' countries of citizenship

```
from rdflib.namespace import RDFS
country_counter = {}
country_labels = set()

for s,p,o in g.triples((None, wdp.P27, None)):
    if o in country_counter:
        country_counter[o] += 1
    else:
        country_counter[o] = 1

    for s1,p1,o1 in g.triples((o, RDFS.label, None)):
        country_labels.add( (s1, o1.strip()))

pprint.pprint(country_counter)
pprint.pprint(country_labels)
```

... and the number of historians for each country.

```
country_counter_labels = {}
for uri,label in country_labels:
    for uri2, count in country_counter.items():
        if uri2 == uri:
          country_counter_labels[label] = count

pprint.pprint(country_counter_labels)
```

Q4

Write a SPARQL query to retrieve in ARTchives all the historians that were born in Berlin (wd:Q64).

```
PREFIX wdt:<http://www.wikidata.org/prop/direct/>
PREFIX wd: <http://www.wikidata.org/entity/>
SELECT DISTINCT ?historian
WHERE {
      ?coll wdt:P170 ?historian.
      ?historian wdt:P19 wd:Q64 .
}
```

Write the SPARQL query to retrieve in Wikidata all the art historians (that are both in ARTchives and Wikidata!) that were born in Germany.

```
SELECT DISTINCT ?historian
WHERE {
 VALUES ?historian {""" + historians + """} .
 ?historian wdt:P19 ?birthplace .
 ?birthplace rdfs:label ?birthplace_label .
 ?birthplace wdt:P17 wd:Q183 .
 FILTER (langMatches(lang(?birthplace_label),
"EN"))
}
```

Q5

Submit your notebook

```
scatt_plot = sns.catplot(x="period_label", y="count_coll", height=2, aspect=5.5, jitter=False,
data=data_by_year)

scatt_plot.set_xticklabels(rotation=90)
```

Q6

Apply the apriori algorithm to calculate which people (art:hasSubjectPerson) mostly co-occur in ARTchives collections. Write here the three rules with the highest confidence in the form: (antecedents) - (consequents)

```
(Alessandro Contini Bonacossi) - (Bernard Berenson)
(Bernard Berenson) - (Alessandro Contini Bonacossi)
(Ernst Kitzinger) - (Erwin Panofsky)
```

You did well!

Results

candidate	Q1	Q1	Q1	Q1	Q1	Q2	Q2	Q2	Q2	Q2	Q2	Q2	Q2	Q2	Q2	Q3	Q3	Q4	Q4	Q5	Q6	AVG
silje.eriksen@studio.unibo.it	9	7	9	8	7	10	10	10	10	10	10	3	10	10	10	10	5			10	10	8.842105263
corrado.consiglio@studio.unibo.it	9	9	9	9	9	10	10	10	10	10	10	10	7	7	10	10	10	10	10	10		9.45
daniele.spedicati@studio.unibo.it						10	10	10	10	10	10	10	9	5	10							9.4
chiara.parravicini@studio.unibo.it						10	10	10	10	10	10	10	9	8	10	10	5	5	5			8.714285714
crosillag@gmail.com						10	10	10	10	5	10	10	9	8	10	10	7	5	5	10	7	8.5
erica.andreose@studio.unibo.it						10	10	10	10	10	10	8	9	10	10	5	5	5	5	10	10	8.5625
anastasiia.marchenk2@studio.unibo.it						10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10
ludi.yang@studio.unibo.it																10	10			5		8.333333333

Thanks!

Do you have any questions?

marilena.daquino2@unibo.it

https://github.com/marilenadaquino/information_visualization